Topo-cluster isolation distributions and photon-energy containment vs inner-R

PPG12 isolated photon analysis

April 18, 2026

1 Motivation

The nominal PPG12 isolation variable is the R = 0.4 topo-cluster energy around the seed cluster: $\mathrm{recoiso}_{ET}^{\mathrm{nom}} = \mathrm{cluster_iso_topo_04}$, which by construction in `CaloAna24` equals $\Sigma_{R<0.4}\, E_T^{\mathrm{topo}} - E_T^{\mathrm{seed}}$. To study an "annulus" variant we subtract an inner ring, parametrised by a new configuration key `iso_topo_innerr`:

$$\mathrm{recoiso}_{ET}^{\mathrm{ann}}(R_{\mathrm{in}}) = \mathrm{cluster_iso_topo_04} \;-\; \mathrm{cluster_iso_topo}_{R_{\mathrm{in}}}.$$

Because both terms have a common $-E_T^{\mathrm{seed}}$, the subtraction amounts to the sum in the annulus $R \in [R_{\mathrm{in}}, 0.4]$. The question addressed here is how much of the photon's own energy sits outside a given inner radius R_{in}, and how the one-dimensional isolation distribution evolves with R in truth-matched signal MC, inclusive MC, and data.

2 Samples and selection

- Truth-matched signal MC: photon5 + photon10 + photon20, cross-section weighted. For a reco cluster to enter the study, its `cluster_truthtrkID` must map to a truth particle with `pid` = 22, `photonclass` = 1 (direct), and `particle_truth_iso_03` < 4 GeV (fiducial).
- Inclusive MC: photon5/10/20 + jet5/8/12/20/30/40, cross-section weighted, no truth match (all reconstructed clusters).
- Data: 77 `part_*_with_bdt_split.root` files from the ana521 production, capped at 3×10^6 clusters for this study.
- *Pre-common*: no cluster-quality selection beyond the sample filter.
- *Post-common*: require `cluster_npb_score` > 0.5 (the only active common cut in `config_bdt_nom.yaml`, since `wr_cogx_bound` = 0 and `cluster_weta_cogx_bound` = 2 disable the shape cuts).

Inner-R branches scanned: $R_{\mathrm{in}} \in \{0.05, 0.075, 0.10, 0.20, 0.30, 0.40\}$.

3 Isolation distributions vs inner radius

Figure 1 overlays the six $\mathtt{cluster_iso_topo}_R$ distributions on each of the three samples. Figure 6 compares, per R, the data and inclusive MC distributions. Table 1 summarises the post-common mean and RMS.

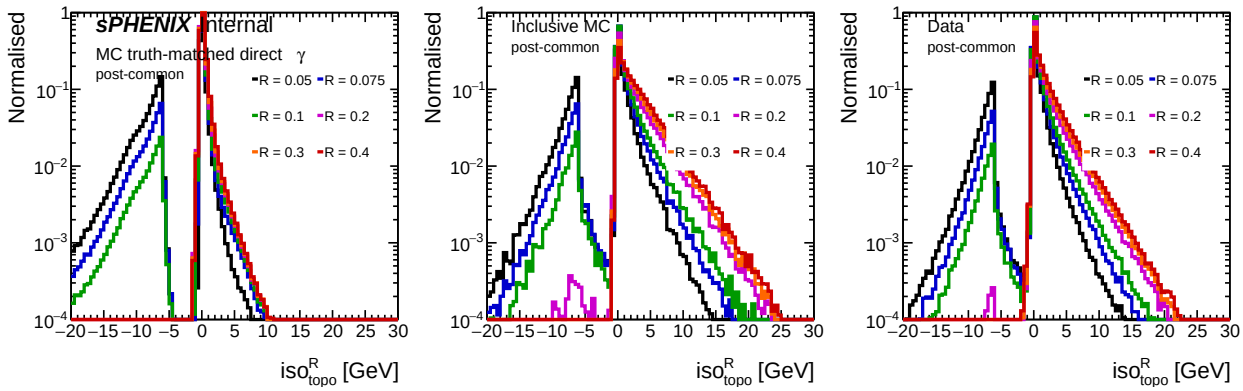

Figure 1: `cluster_iso_topo`$_R$ distributions (log-y, unit-normalised) for the six inner radii, on truth-matched signal MC (left), inclusive MC (middle) and data (right), all post-common. Inclusive in cluster E_T.

3.1 Per-cluster-E_T expansion (linear y)

Figure 2 to 5 repeat Figure 1 in four cluster-E_T windows with linear y-scale to make the peak structure visible. All samples and the six inner radii follow the same colour assignment; post-common selection applies throughout.

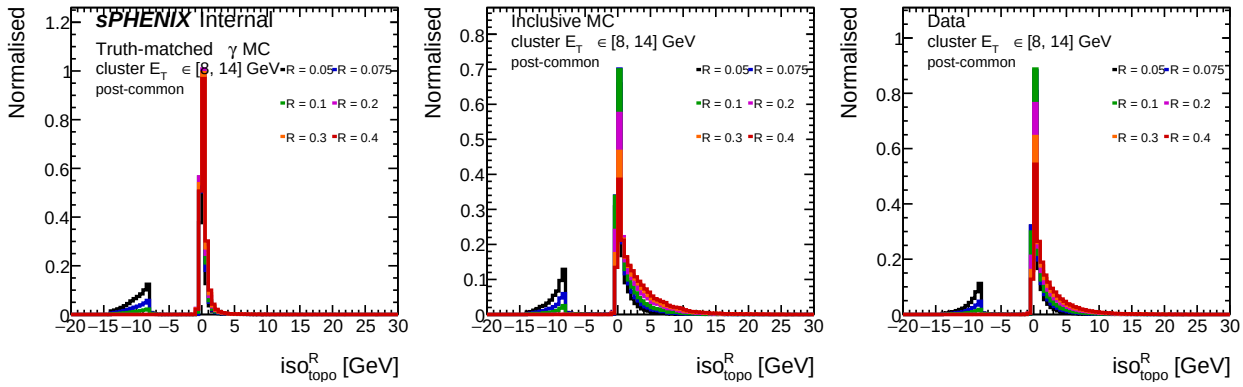

Figure 2: Cluster $E_T \in [8, 14]$ GeV, linear y.

R	MC truth-matched photons		Inclusive MC		Data	
	mean	RMS	mean	RMS	mean	RMS
0.05	−2.83	4.42	−1.72	4.13	−1.27	3.75
0.075	−1.19	3.39	−0.19	3.43	−0.07	3.37
0.10	−0.33	2.29	+0.72	2.87	+0.59	3.70
0.20	+0.24	0.68	+1.98	2.82	+1.49	8.42
0.30	+0.27	0.69	+2.50	3.08	+1.97	17.2
0.40	+0.31	0.71	+2.88	3.24	+2.39	27.1

Table 1: Mean and RMS (GeV) of `cluster_iso_topo`$_R$ in the three samples, post-common. All values are cross-section weighted for the MC rows.

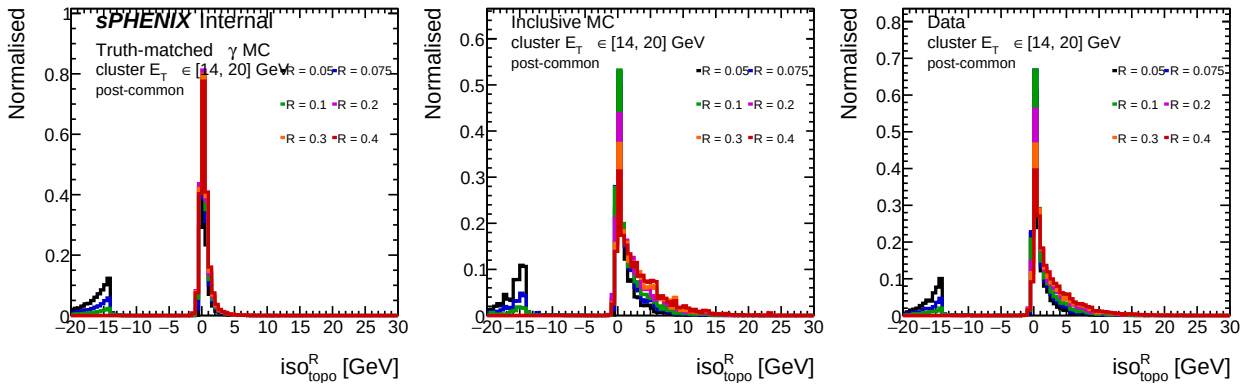

Figure 3: Cluster $E_T \in [14, 20]$ GeV, linear y.

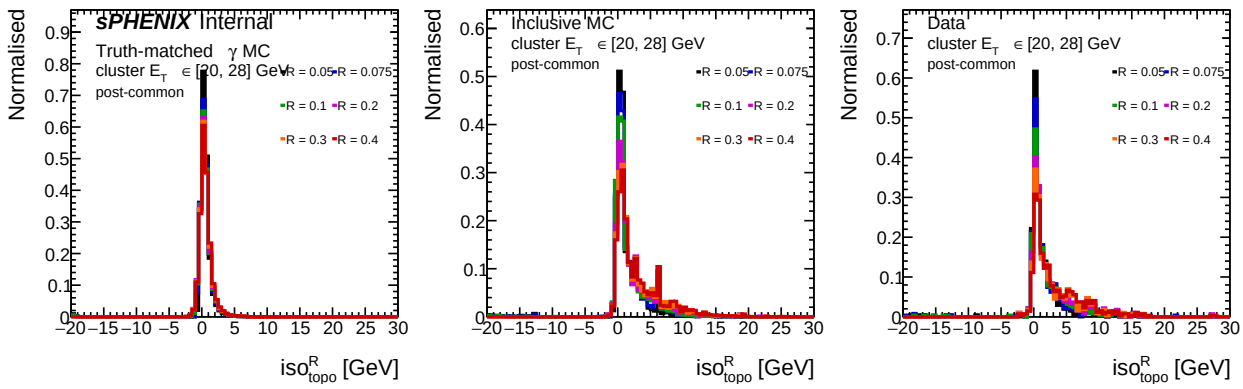

Figure 4: Cluster $E_T \in [20, 28]$ GeV, linear y.

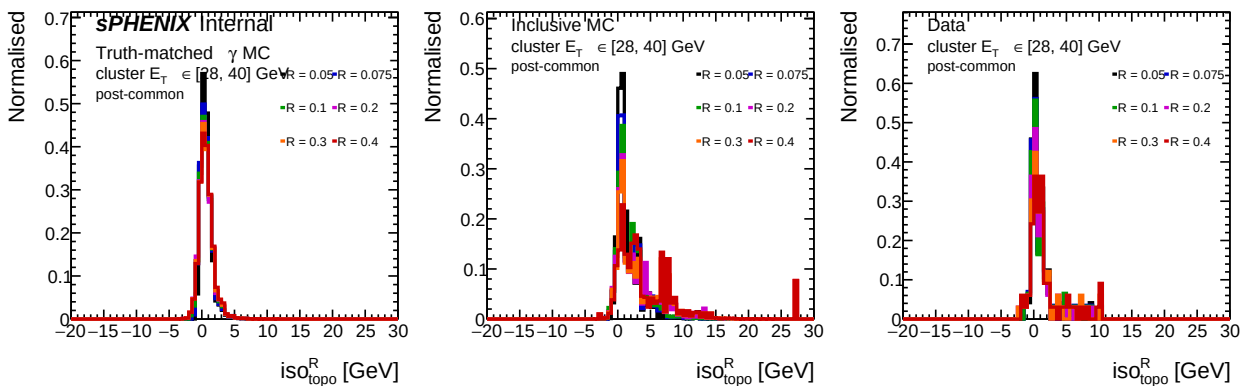

Figure 5: Cluster $E_T \in [28, 40]$ GeV, linear y.

Observations. Across all three samples the isolation distribution tightens as $R \to 0$; the signal-MC RMS shrinks from 0.71 GeV at R = 0.4 to 0.68 GeV at R = 0.2 and grows to 4.4 GeV at R = 0.05 because most of the photon's energy still sits inside this tiny cone. For data, the RMS at R = 0.4 reaches 27 GeV, driven by long tails from high-E_T jets that overlap with the seed cluster — a feature the inclusive-MC mix (RMS 3.2 GeV at R = 0.4) does not reproduce. The mean in inclusive

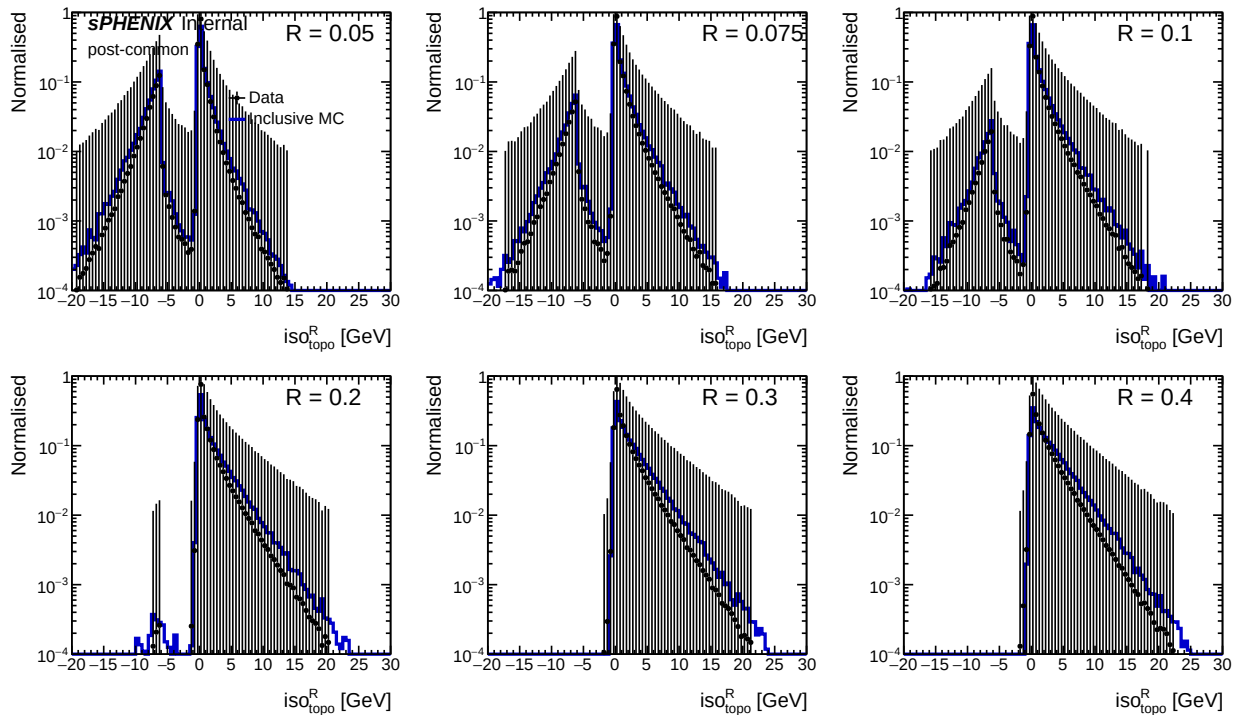

Figure 6: Post-common `cluster_iso_topo`$_R$ distribution, one panel per inner radius. Data (black markers) vs inclusive MC (blue line), both unit-normalised.

MC is systematically ≈ 0.5 GeV above data at every R, consistent with the MC mix being slightly jet-richer than the as-taken data sample.

4 Photon self-energy containment vs inner radius

For truth-matched direct photons the containment fraction is defined as

$$f_{\text{contain}}(R) \;=\; \frac{\text{cluster_iso_topo}_R \;+\; E_T^{\text{cluster}}}{p_T^{\gamma,\text{truth}}} \;\approx\; \frac{\Sigma_R\, E_T^{\text{topo}}}{p_T^{\gamma,\text{truth}}} \tag{1}$$

(the $-E_T^{\text{seed}}$ in the iso definition and the $+E_T^{\text{cluster}}$ in the numerator cancel up to clustering residuals, leaving the in-cone energy over truth p_T). For isolated photons, $f_{\text{contain}}(R)$ is the fraction of photon energy within R of the cluster, and the leakage $1 - f_{\text{contain}}(R)$ is the fraction that falls outside. Figure 7 shows $\langle f_{\text{contain}}(R)\rangle$ for the inclusive signal and for four truth-p_T bins.

Observations. Containment reaches $\sim$95% at $R = 0.10$ and saturates above unity by $R = 0.20$ (the slight excess over 1.0 reflects residual background plus cluster-shape calibration details). The pT dependence is mild: at $R = 0.10$ the average contained fraction drops from 92% in $[8, 14]$ GeV to 89% in $[28, 40]$ GeV, a $\sim$3 percentage-point broadening consistent with increased lateral spread at higher energies.

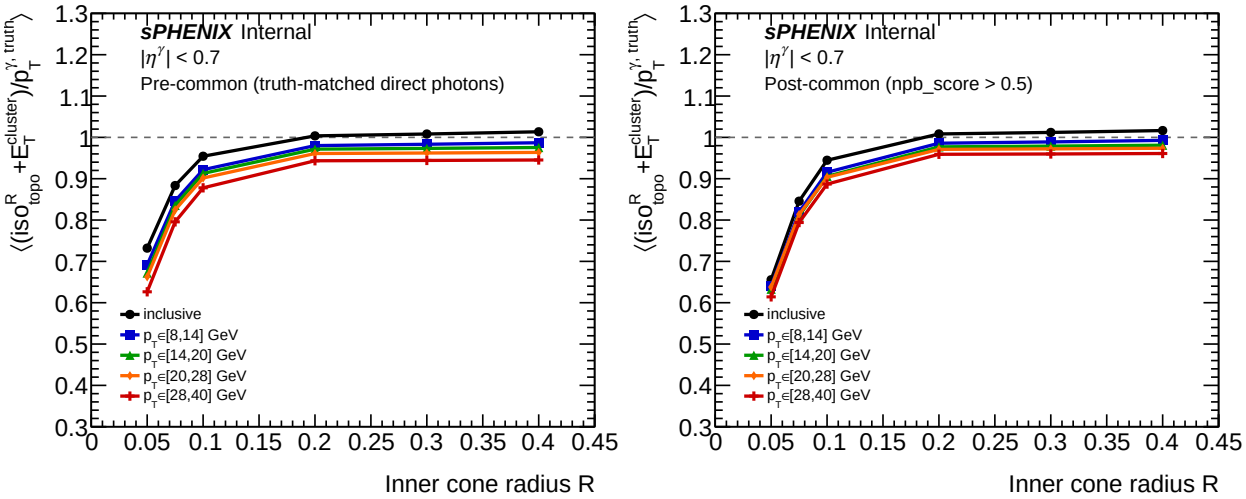

Figure 7: Mean containment $\langle f_{\text{contain}}(R)\rangle$ in truth-matched direct photon signal MC, pre-common (left) and post-common (right), for the inclusive sample and four truth-p_T bins.

Truth p_T bin	$R = 0.05$	$R = 0.075$	$R = 0.10$	$R = 0.20$	$R = 0.30$	$R = 0.40$
inclusive	0.656	0.846	0.945	1.008	1.012	1.016
$[8, 14]$ GeV	0.640	0.820	0.916	0.986	0.989	0.993
$[14, 20]$ GeV	0.633	0.810	0.906	0.977	0.979	0.981
$[20, 28]$ GeV	0.638	0.813	0.903	0.971	0.972	0.974
$[28, 40]$ GeV	0.614	0.794	0.887	0.959	0.960	0.961

Table 2: Mean containment $\langle f_{\text{contain}}(R)\rangle$, post-common, by truth photon p_T bin. Small R under-contains the photon; higher p_T marginally broadens the shower profile so containment decreases with p_T at fixed R.

5 Implication for the inner-R scan

At $R_{\text{in}} = 0.05$ the inner ring removes only $\sim 66\%$ of the photon's energy while the outer $\mathrm{R} = 0.4$ cone contains essentially all of it; subtracting the former from the latter therefore leaves $\approx 34\%$ of the photon in the recoiso$_{ET}$ definition. That $\sim 4\,\text{GeV}$ contamination at $p_T^\gamma \sim 10\,\text{GeV}$ shifts the observed `cluster_iso_topo_04` − `cluster_iso_topo_005` mean from near zero to $+3.9\,\text{GeV}$, which is what was seen in the inner-R scan of the purity MC closure and explains why the ABCD iso region collapses at that working point. Inner radii of $R_{\text{in}} \geq 0.10$ leave $\lesssim 5\%$ photon residue and are the physics-motivated range for an annulus-isolation study if a clean photon-independent isolation variable is desired.

6 Files

- Analysis: `efficiencytool/iso_topo_containment_study.py`, `efficiencytool/iso_topo_make_rootfile.`
- Histograms: `efficiencytool/results/iso_topo_containment.root`.
- Arrays: `efficiencytool/results/iso_topo_containment.npz`.
- Plot macro: `plotting/plot_iso_topo_containment.C`.

- Figures: `plotting/figures/iso_topo_{dist_overlay,dist_data_vs_mc,containment_vs_R}.pdf`.